

Dataset Sources and Access

- **New York City (NYC) DOHMH Food Inspections**
 - Source: [NYC Open Data Portal](#)
 - API: <https://data.cityofnewyork.us/api/v3/views/43nn-pn8j/query.json>
 - Coverage: All sustained or pending violation citations from inspections of restaurants and college cafeterias up to three years before the data pull.
 - Licensing/Terms: Publicly available through the NYC Open Data Portal; users agree to NYC.gov Terms of Use and Privacy Policy.
- **Chicago Food Inspections**
 - Source: [Chicago Data Portal](#)
 - API: <https://data.cityofchicago.org/api/v3/views/4ijn-s7e5/query.json>
 - Coverage: Inspections of restaurants and other food establishments from January 1, 2010, to present. Inspections are done by CDPH and reviewed by State-licensed Environmental Health Practitioners.
 - Licensing/Terms: Publicly available through the Chicago Data Portal; no guarantee on content, accuracy, or completeness.

Data Cleaning and Integration

- Outliers and missing/irrelevant values removed:
 - Chicago: Inspections labeled Business Not Located, Not Ready, No Entry, or Out of Business.
 - NYC: Inspections with Critical Flag = Not Applicable.
 - Establishments with an inspection date of 1/1/1900 (not yet inspected) were excluded.
- Column standardization for integration:

NYC columns renamed to match the Chicago dataset schema:

```
{"DBA": "DBA Name", "BORO": "Boro", "STREET": "Street", "ZIPCODE": "Zip",  
"CUISINE DESCRIPTION": "Cuisine Description", "INSPECTION DATE": "Inspection Date",  
"VIOLATION DESCRIPTION": "Violations", "CRITICAL FLAG": "Critical Flag"}
```

- String fields such as DBA Name and Violations were converted to uppercase for consistency.

- Random sampling was used to reduce the dataset size (75% of observations, random_state=477) to fit GitHub constraints.

Selected Attributes for Integrated Dataset

Column	Type	Description
RecordID	Int	Unique identifier for each observation
DBA Name	str	Legal/public business name of the establishment
CAMIS	Int64	Unique NYC inspection ID (NYC only)
License Number	Int64	Unique establishment license number (Chicago only)
Address	str	Full address of the establishment
Street	str	Street name (NYC only)
City	str	Chicago or New York City
Boro	str	Borough (NYC only)
State	str	State of establishment (IL or NY)
Zip	Int	Zip code
Inspection Date	datetime	Date of inspection
Risk	str	Risk category (1–3; Chicago only)
Critical Flag	str	Indicates critical violation (Critical / Not Critical / Not Applicable; NYC only)
Facility Type	str	Type of establishment (e.g., restaurant, grocery, bakery)
Cuisine Description	str	Cuisine type (optional)
Violations	str	Description of violations noted during inspection
Results	str	Inspection outcome (Pass / Pass with Conditions / Fail / Not Located / Out of Business; Chicago only)

Data Quality

- Qualitative inconsistencies addressed: street name formats, abbreviations, and typos.
- Quantitative issues addressed: non-numeric values in **License** and **Zip**, missing dates.
- Duplicates minimized, though Chicago documentation notes some may remain.
- Datasets deemed suitable for analysis after cleaning.

Storage & Versioning

- Original CSV files stored on Google Drive for asynchronous access due to file size limits (>100 MB).
- SHA-256 hashes computed to verify dataset integrity.
- Edit histories from OpenRefine stored in a GitHub repository for reproducibility.

Licenses for Data & Software

Field / Artifact	Type	Description / Notes	License / Terms of Use
<i>Python</i>	Software	Programming language used for analysis, data cleaning, and visualization	Python Software Foundation License Version 2
<i>NumPy</i>	Software	Library for numerical computation and array operations	BSD-style, free to use, modify, distribute, disclaimers apply
<i>Pandas</i>	Software	Library for data manipulation, analysis, and cleaning	BSD-style, free to use, modify, distribute, and must retain the copyright notice

<i>Matplotlib</i>	Software	Library for creating visualizations and plots	BSD-style, free to use, modify, distribute
<i>NYC Food Inspections</i>	Dataset	Contains all sustained and pending violation citations for restaurants and college cafeterias in NYC	Publicly available; usage must comply with NYC.gov Terms of Use & Privacy Policy
<i>Chicago Food Inspections</i>	Dataset	Inspections for restaurants and food establishments in Chicago from 2010 to the present	Publicly available; usage must comply with the Chicago Data Portal Terms of Use; no warranty